



Designing for More People Near Frequent Buses

March 2026

Motivation for the problem taken up

- Usage of public transportation is critical to help solve congestion in Bangalore
- Expecting people to jump through hoops to use public transit is unrealistic
- PT needs to be easily accessible (within say 5 mins walk) and frequency needs to be reasonable (not more than a 10 min wait) for people to consider it
- We took two areas not too far from each other (BTM and HSR) to check
 - How do they compare with each other - BTM is a much older part of Bangalore
 - How well do they perform on PT (distance and frequency)
 - What are the opportunities to improve PT performance in these areas

Problem Statement

- Assess and compare the proximity to frequent bus transport (every 10 minutes, within 500m distance) in two areas of Bengaluru and suggest additional routes for coverage
- Datasets used: Vonter/bmtc-gtfs (for bus stops and routes); Open City 2025-Bengaluru population, Static bus timings
- Analysis methods: Heat-mapping, mapping population density and land use
- Assumptions made:
 - Analysis considers stated frequency and not actual frequency

Research Questions

Primary question:

- How is the proximity to frequent buses in HSR Layout compared to BTM Layout within a 500m walking distance and how can it be improved?

Secondary questions:

- How much of HSR Layout and BTM Layout are serviced by frequent buses within a 500m walking distance?
- What is the extent of bus stops serviced by frequent bus routes?
- What locations/ destinations do the frequent bus routes service?

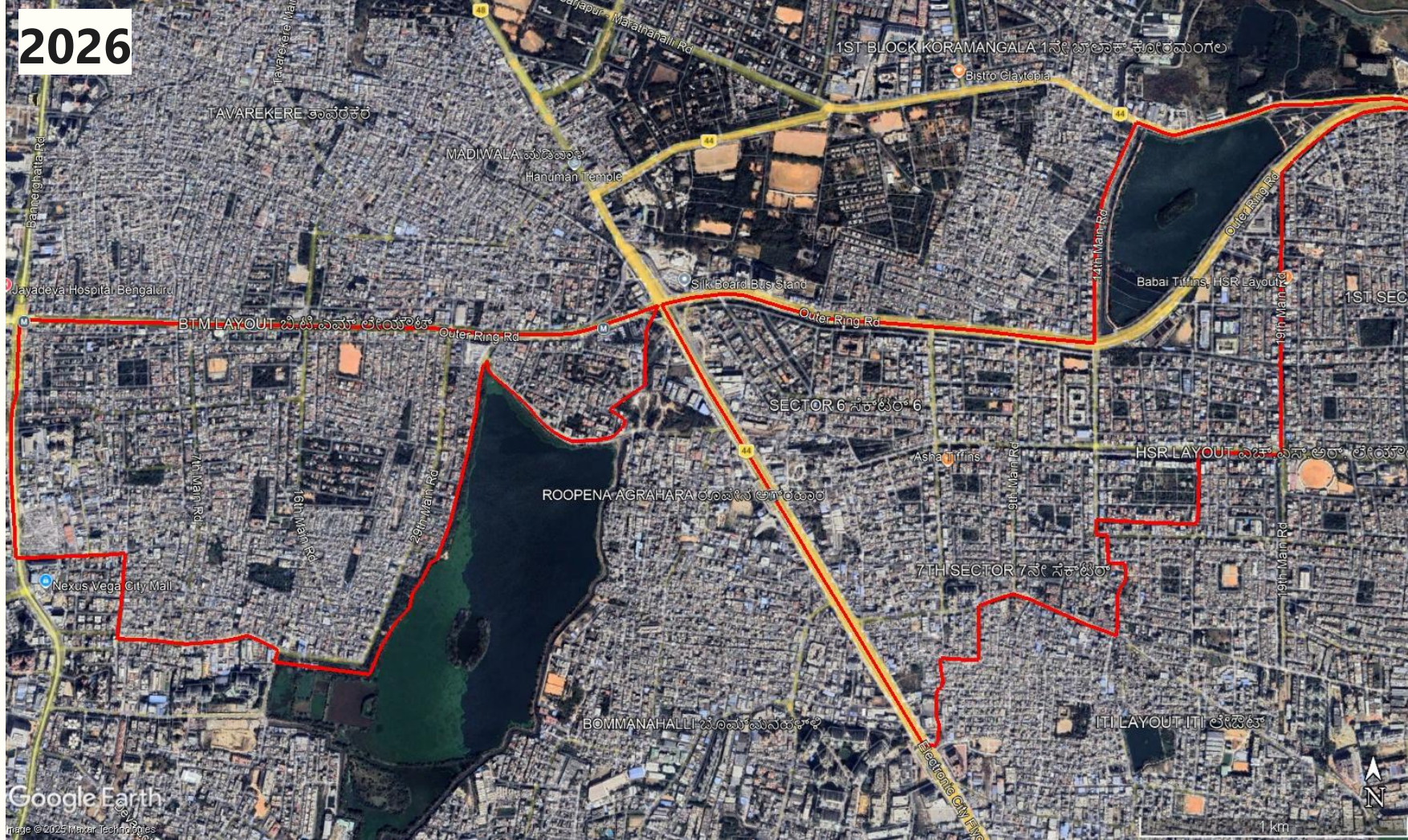
2011



Google Earth

Image © 2025 Maxar Technologies

2026

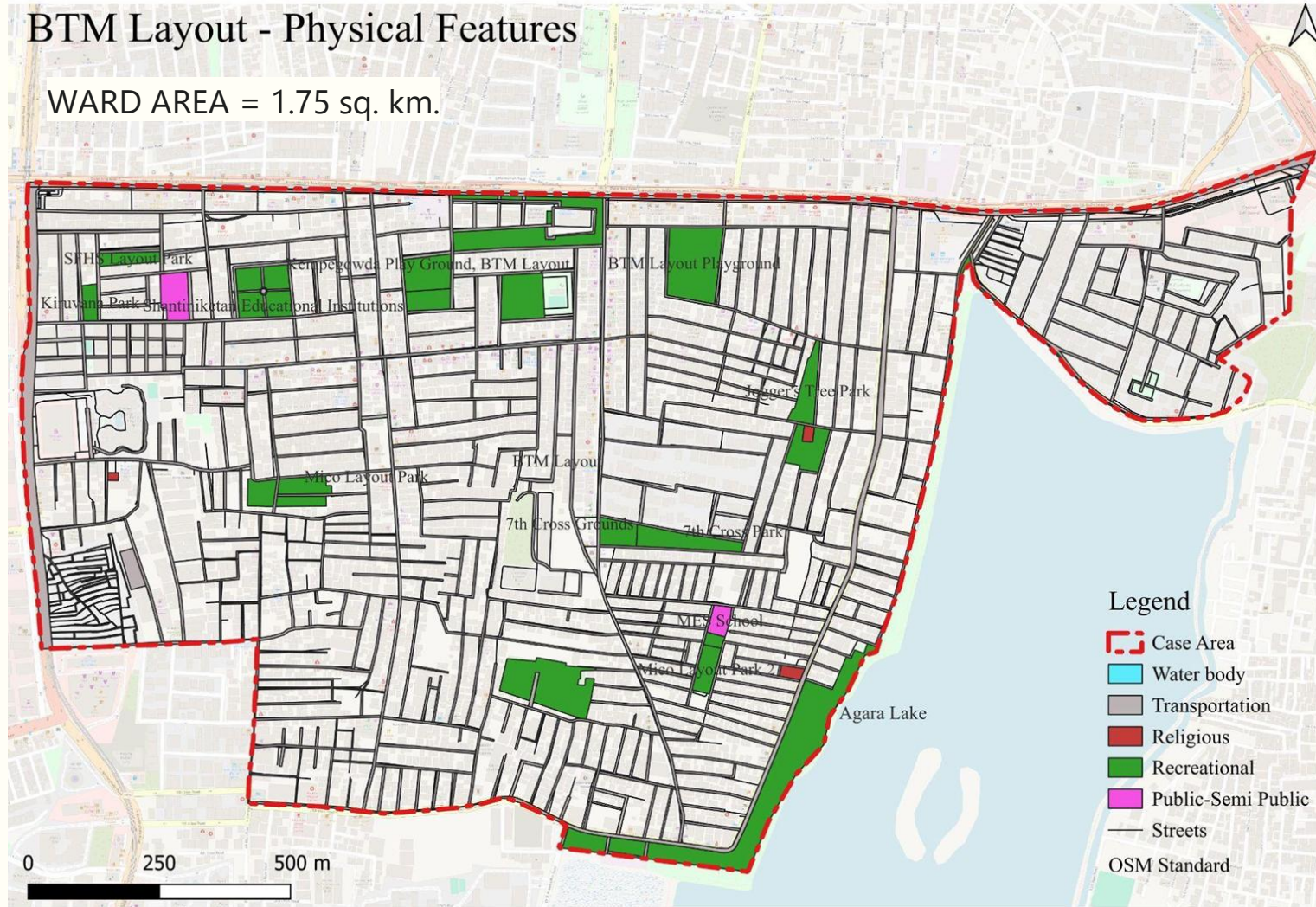


Google Earth

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BTM Layout - Physical Features

WARD AREA = 1.75 sq. km.



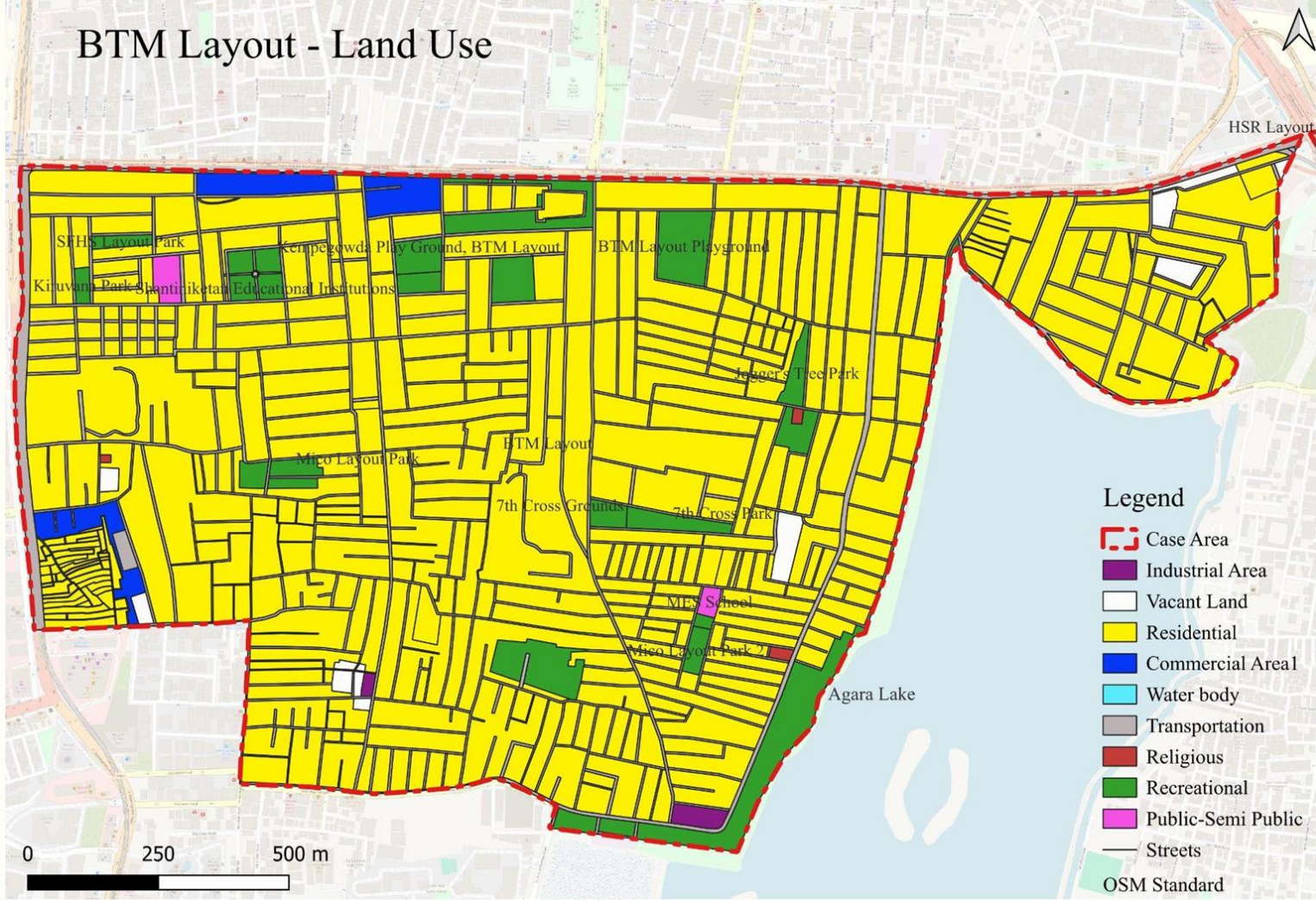
BTM Layout: Characteristics



BTM Layout (development since the 1977)

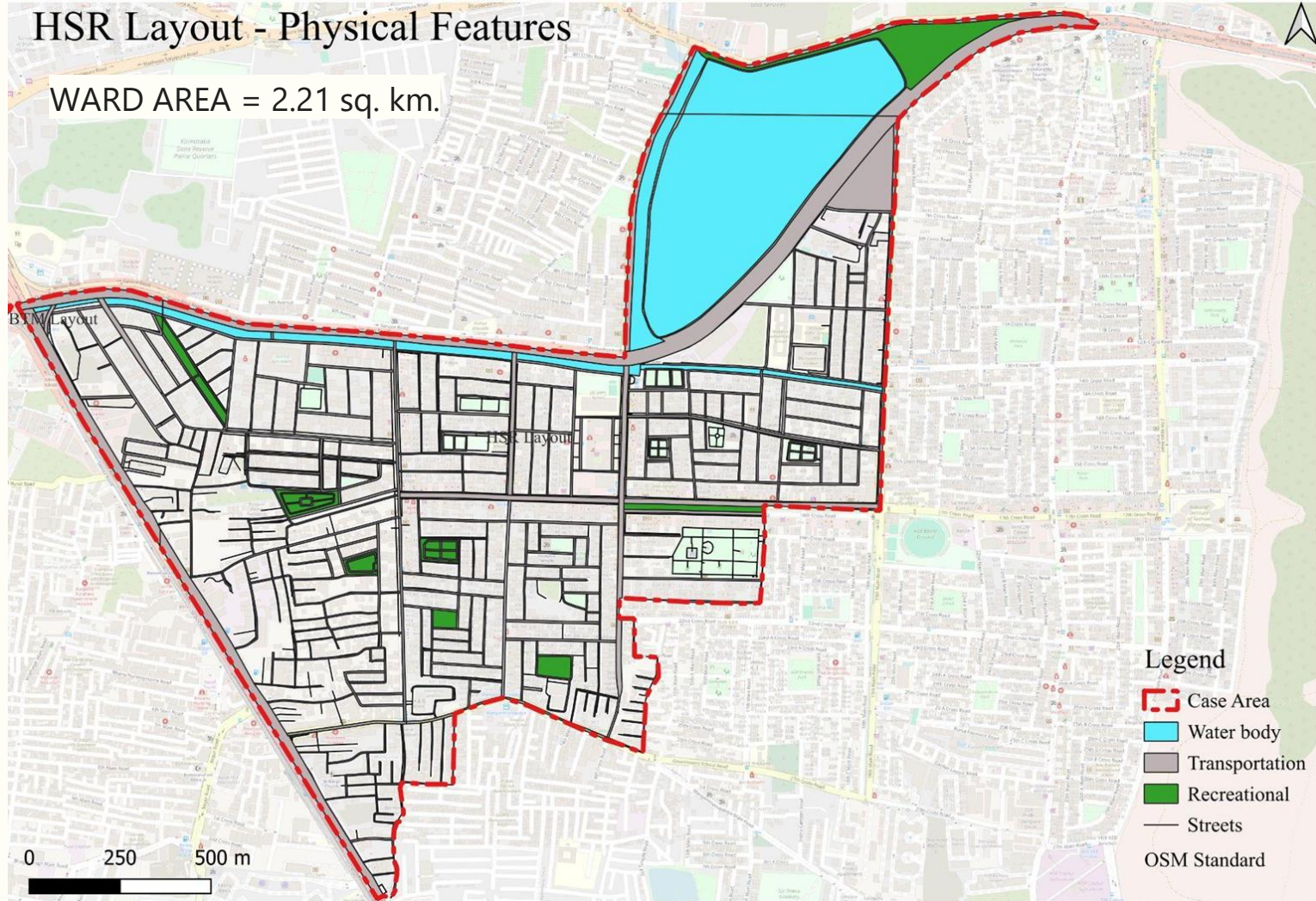
- Prime South-Bengaluru locality, known for its high-density, mid-range housing, excellent connectivity to IT hubs (Koramangala, HSR Layout, Electronic City)
- Vibrant, youth-centric cafes and nightlife
- Independent houses, older apartment complexes, and popular PGs for students and working professionals
- Popular among young professionals and students due to its central location and, rental affordability compared to neighboring HSR and Koramangala

BTM Layout - Land Use



HSR Layout - Physical Features

WARD AREA = 2.21 sq. km.



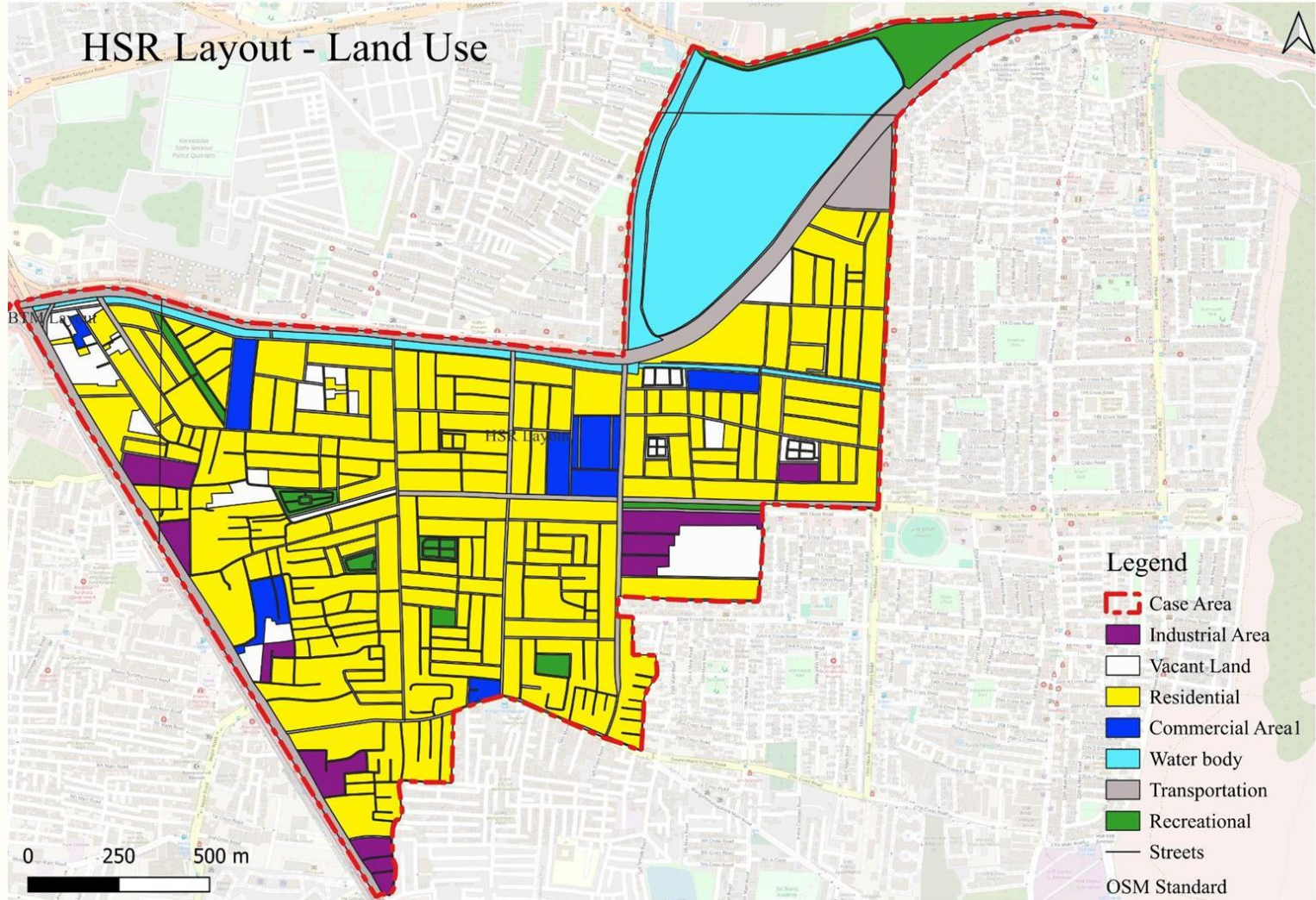
HSR Layout: Characteristics



HSR Layout (rapid development since the 2000s)

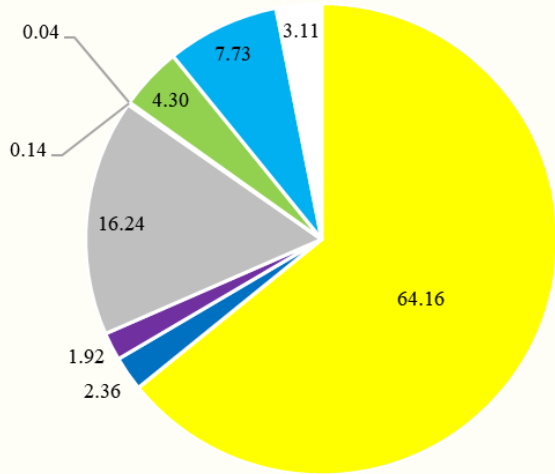
- Premier, well-planned suburb in South Bengaluru, known for its wide, tree-lined roads, 25+ parks, and as a major hub for startups
- Designed with a grid system, the area is characterised by wide roads, generally good infrastructure, and 25+ green spaces.
- Area features 15+ top schools (e.g., NPS, VIBGYOR), high-end fitness centers, and numerous cafes and restaurants.

HSR Layout - Land Use



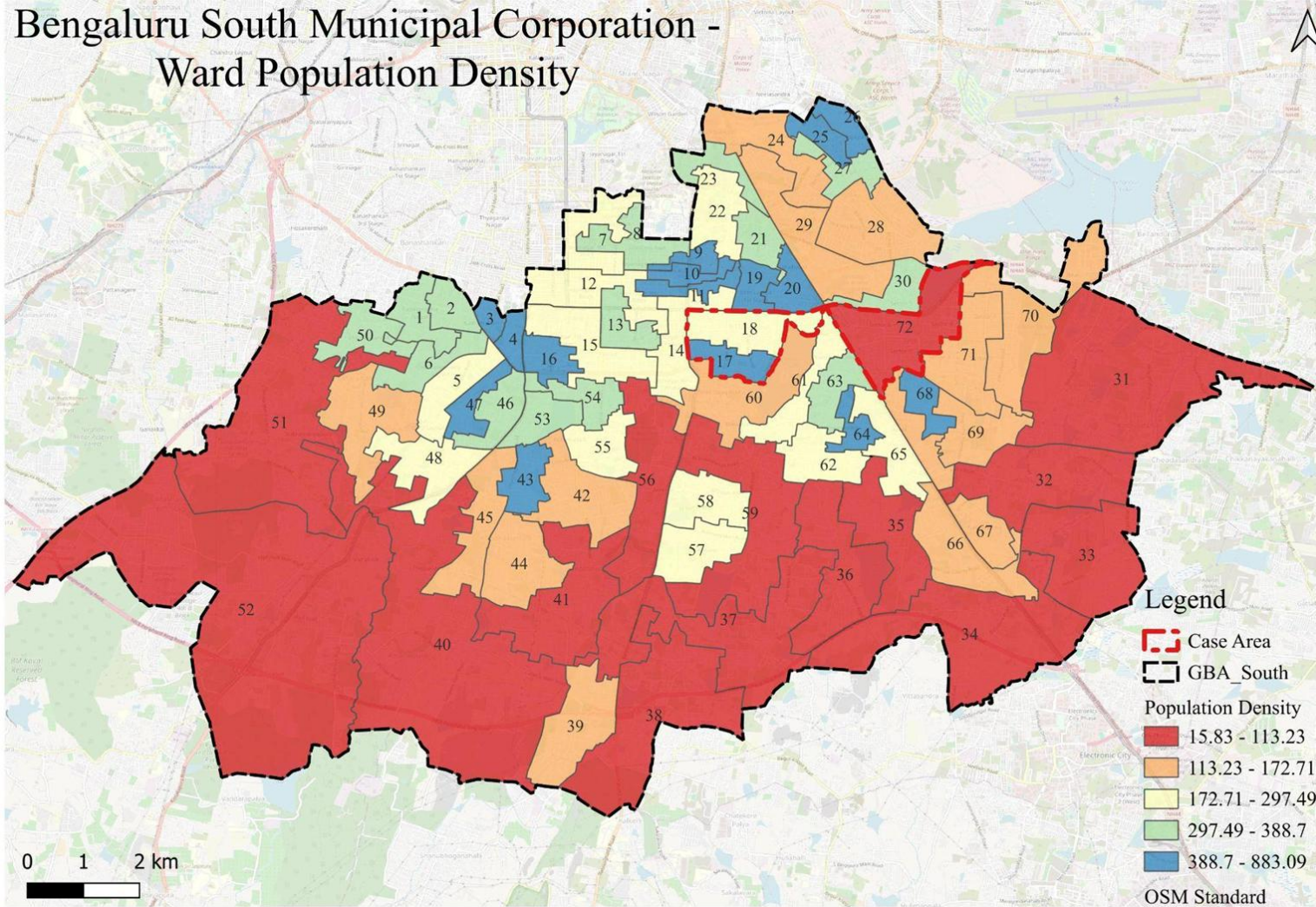
Land use Analysis

Land use %



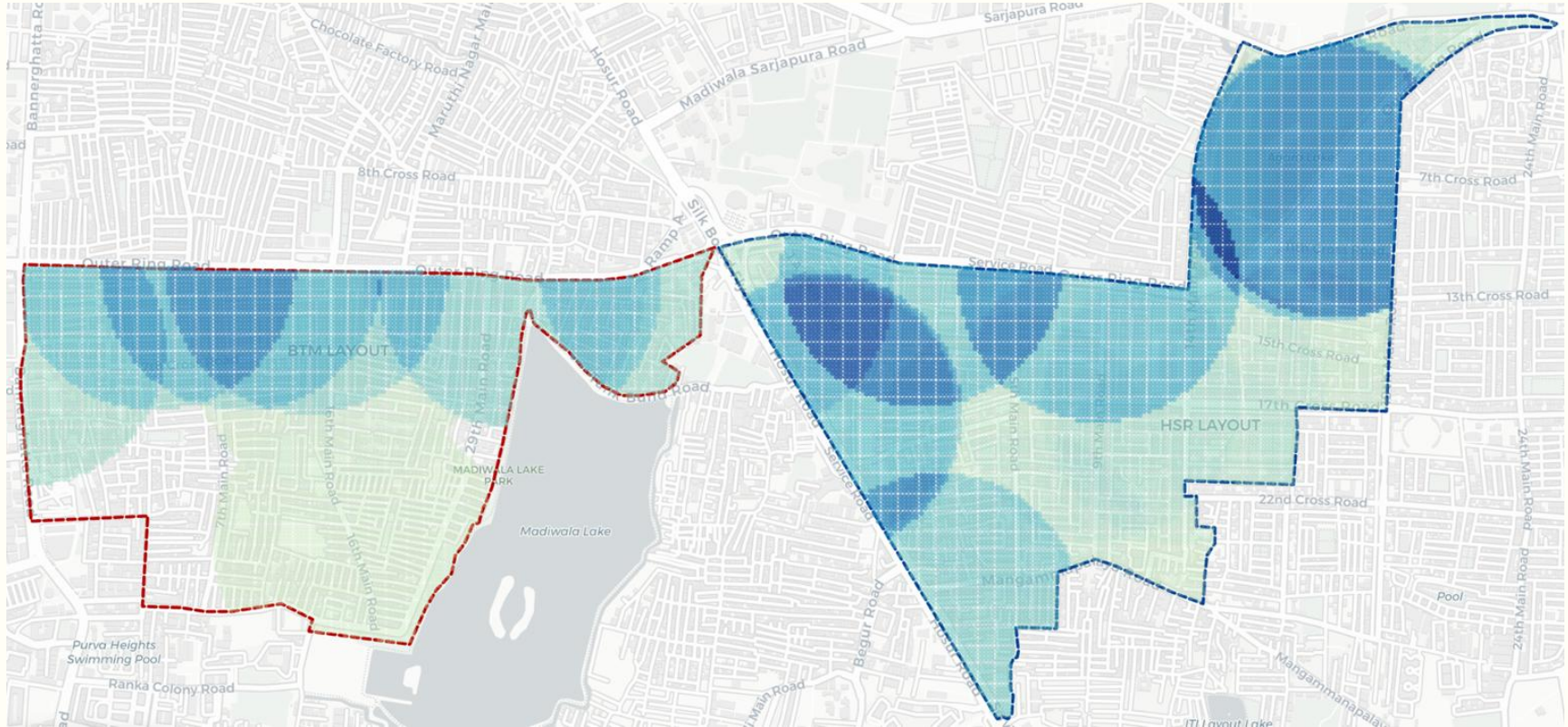
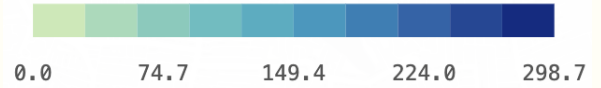
Land use	Area (sq.m.)	Percentage
Residential	261.86	64.16
Commercial	9.65	2.36
Industrial	7.82	1.92
Transportation	66.3	16.24
Public Semi-Public	0.56	0.14
Religious	0.16	0.04
Recreational	17.55	4.30
Water body	31.56	7.73
Vacant	12.68	3.11
Total	408.14	100.00

Bengaluru South Municipal Corporation - Ward Population Density



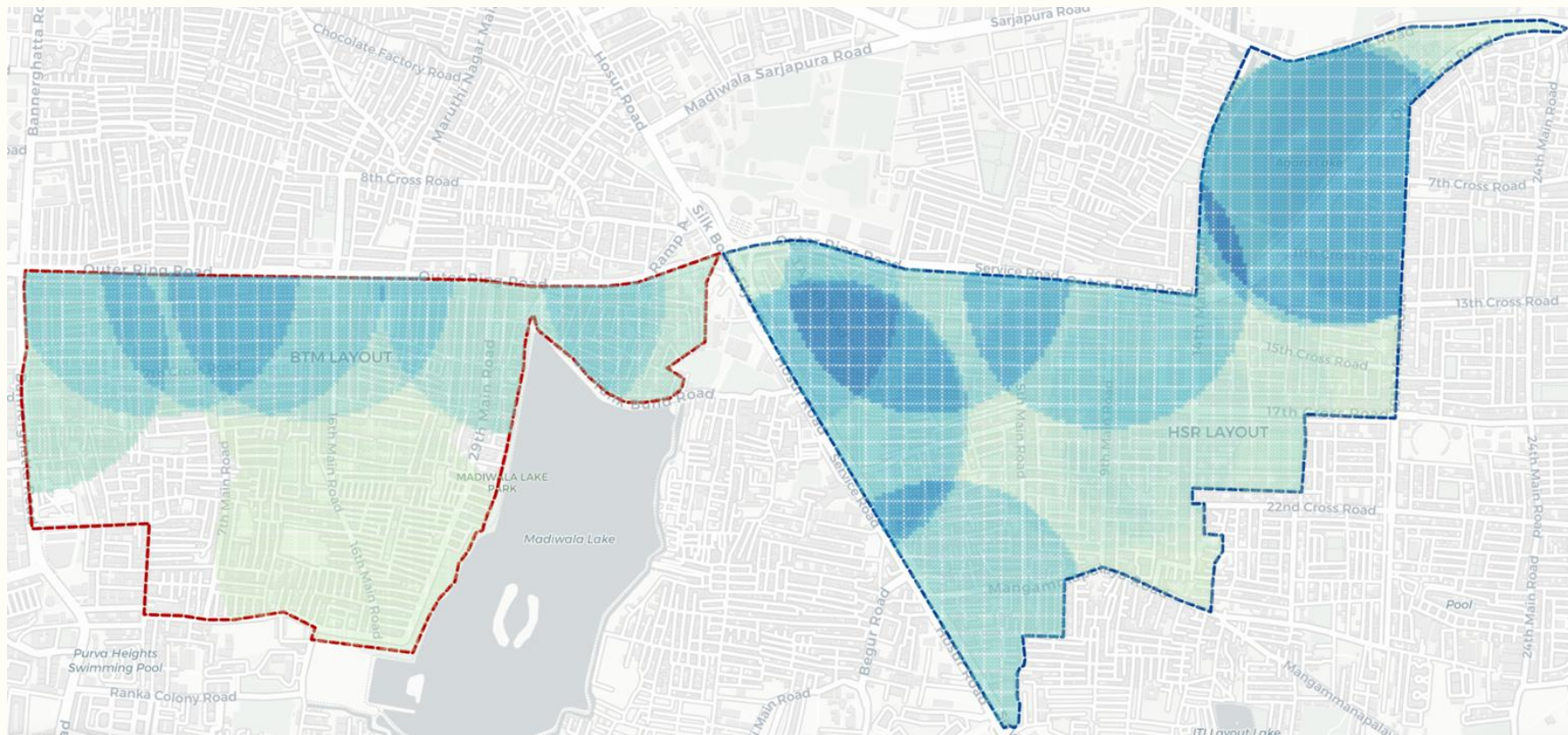
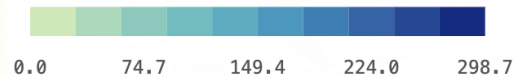
Morning slot

TRIPS / HR WITHIN 500 m



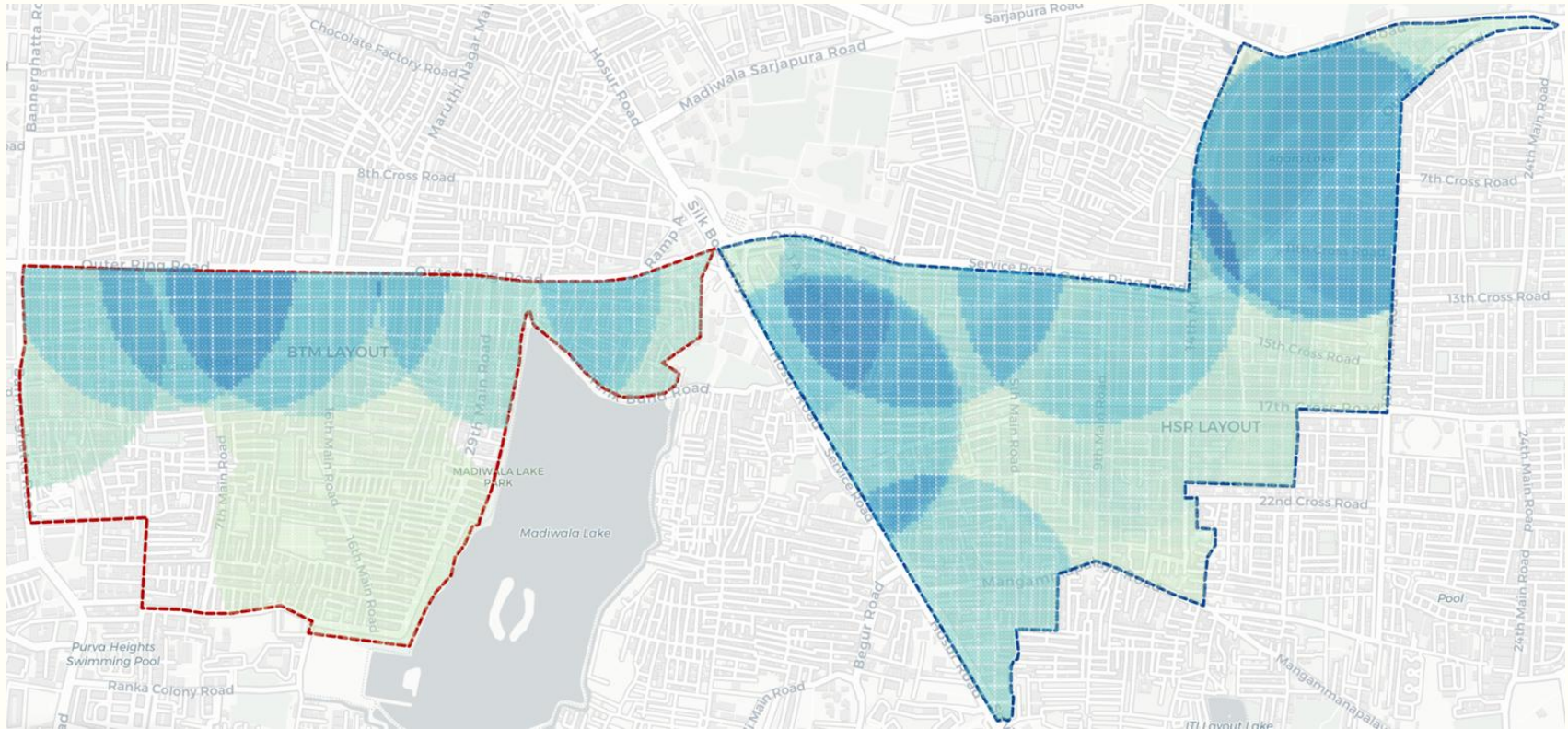
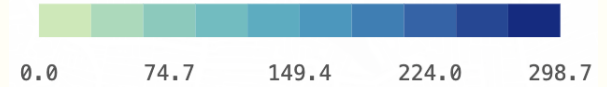
Afternoon slot

TRIPS / HR WITHIN 500 m



Evening slot

TRIPS / HR WITHIN 500 m



Recommendations for HSR connectivity

- Direct connections from major intra-city terminals such as Majestic, Shivajinagar, Banashankari and Marathalli into HSR layout.
- Continuing the use of 9-metre or 7m mini-buses that can navigate the narrower interior roads (e.g., 17th Cross, 27th Main) more easily than standard 12m buses.
- Ensuring the loop comprehensively reaches all four sides of the layout, including Silk Board, Mangammanapalya, and Somasundrapalya, rather than primarily focusing on the Agara side.

Team

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